INCH-POUND

A-A-59755
5 November 2004

COMMERCIAL ITEM DESCRIPTION

WATTMETER, RADIO FREQUENCY

The General Services Administration has authorized the use of this Commercial Item Description (CID) for all federal agencies.

1. SCOPE. This Commercial Item Description (CID), describes a Radio Frequency (RF) Wattmeter. This CID is meant as a minimum requirement for the RF Wattmeter in which only those manufacturers that meet or surpass the following requirements are to supply this wattmeter per this CID. The RF wattmeter described herein is a portable, insertion-type, RF wattmeter used to measure continuous wave (CW) forward and reflected power in 50 ohm transmission lines using removable power elements. The RF Wattmeter described herein has a detachable load resistor accessory for terminated RF power measurements.

2. CLASSIFICATION. Equipment covered by this CID shall be commercially available equipment and may be modified to the extent necessary to meet this description. The equipment shall be Class 3 in accordance with MIL-PRF-28800, except as specified herein.

3. SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

3.1 Safety and Environmental. The equipment shall meet all safety and environmental requirements as specified in MIL-PRF-28800 for the classification as stated herein.

3.1.1 Temperature Operating. The equipment shall conform to its specified performance and accuracy while being operated in a temperature range of +10°C to +45°C.

3.1.2 Temperature Non-operating. The equipment shall conform to its specified performance and accuracy after being stored in an environment in the temperature range of -10°C to +45°C.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611

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3.2 Electrical Power Sources. The equipment shall operate from nominal single-phase commercial, military, and shipboard power sources of 120 volts AC (±10%) or 240 volts AC (±10%) at line frequencies of 47.5 Hz to 52.5 Hz and 57 Hz to 63 Hz. Transient state conditions shall meet MIL-PRF-28800 requirements for the classification specified herein. The equipment shall also have internal rechargeable battery power. The battery/batteries shall provide power for the unit for a minimum of 8 hours with average use before recharging is necessary. The batteries shall be capable of being recharged to full potential in less than 16 hours.

3.3 Reliability. The design of the equipment shall be such that under normal use and operation the equipment does not fail within 1500 hours of operation with a statistical certainty of 95%.

3.4 Calibration and Maintenance Adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. The calibration adjustments, wherever possible, shall be accessible without removal of the equipment case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2000 hours. No special tools shall be required for alignment or adjustment except those furnished with the equipment.

3.5 Performance Requirements. The equipment shall meet all requirements specified herein, after the warm-up period specified in MIL-PRF-28800, under all combinations of input power conditions, output signal conditions and operating service condition as specified in MIL-PRF-28800.

3.5.1 Frequency. The wattmeter frequency range shall include 450 kHz to 2.3 GHz.

3.5.2 Power. The power range of the wattmeter shall be a minimum of 100 milliwatts to 2.0 kilowatts as a directional wattmeter and 0 to 100 watts with a detachable load resistor.

3.5.2.1 Power Accuracy. The accuracy of the wattmeter shall be within ± 7% of full scale for continuous wave (CW) power measurements and within ±8% of full scale for peak envelope power (PEP) power measurement.

3.5.3 Power Elements. The directional wattmeter shall be compatible with standard forward and reflected power Plug-in Elements which correspond to the frequency and power levels to be measured. However, the wattmeter shall not be supplied with the power elements.

3.5.4 Modulation Frequency. The minimum modulation frequency range for the wattmeter shall be 30 Hz to 10 kHz.

3.5.4.1 Percent Modulation. The wattmeter shall measure percent modulation, for power levels greater than one third of full scale, with an accuracy of ±5 % for 0 to 90 % modulation and ±10 % accuracy for 90 to 100 % modulation.

3.5.5 Standing Wave Ratio (SWR). The wattmeter shall measure SWR for the range 1.00 to 85.00 to an accuracy of ±10 % (worst case) of the reading.

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3.5.6 Return Loss. The wattmeter shall measure the return loss of signals for the range 0 to 30.0 dB with an accuracy of ±0.3 dB to corresponding SWR.

3.5.7 Interface.

3.5.7.1 Wattmeter Control. The controls that operate the wattmeter shall be on the front or face panel.

3.5.7.2 Connectors. The wattmeter shall have type N female connectors.

3.5.7.2.1 Impedance. The wattmeter shall have an impedance of 50 ohms.

3.5.7.3 Insertion VSWR. The maximum insertion VSWR of the wattmeter shall be less than or equal to 1.05:1 for frequencies less than 1.0 GHz.

3.5.7.4 Display. The wattmeter shall have a minimum of three (3) character display with floating decimal point. The wattmeter shall be able to display the following operation modes and indicators: Forward power; Reflected power; SWR or VSWR; Return Loss; Percent Modulation; Watts and dBm (dB); Over-range Indicator; and Low Battery.

3.5.7.5 External Interface. The equipment shall be compatible with optional IEEE-488 or RS-232 capability.

3.5.7.6 Pulse Parameters. The equipment shall make measurements with the following minimum square pulse modulation characteristics.

3.5.7.6.1 Pulse Width. The minimum pulse width shall be 1.0 microsecond for 100 MHz to 2300MHz, 2.0 microseconds for 26 MHz to 99 MHz, and 20 microseconds for 2 MHz to 25 MHz.

3.5.7.6.2 Pulse Repetition Rate. The minimum pulse repetition rate shall be 30 pulses per second.

3.5.7.6.3 Pulse Duty Cycle. The minimum duty cycle shall be 0.01%.

3.5.8 Accessories. The equipment shall include the following accessories or equivalent capabilities.

3.5.8.1 Transit Case. The equipment shall include a transit case capable of storing the RF wattmeter, accessories, power cord, and manuals.

3.5.8.2 Load Resistor. The equipment shall include a load resistor accessory. The load resistor shall be air cooled, 100 watts continuous duty, and 50 ohms impedance. The VSWR shall be no more than 1.10:1 from DC to 1.0 GHz and no more than 1.25:1 from 1.0 GHz to 2.3 GHz.

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3.5.9 Physical Characteristics. The size and weight of the equipment shall be consistent with current commercial practices. The equipment shall be portable with the following physical characteristics.

3.5.9.1 Dimensions. The height, width, and depth of the equipment shall be compatible with hand-held or portable operation. The equipment shall be within 5 inches high, 8 inches wide, and 10 inches deep.

3.5.9.2 Weight. The weight of the equipment shall be compatible with hand-held or portable operation. The total weight of the equipment, excluding accessories and manuals, shall not exceed 11 pounds (5 kilograms).

3.5.10 Manuals. The equipment shall be delivered with operator, maintenance, illustrated parts, and calibration manuals. Format and quantity shall be as specified in the contract or order. Level of maintenance philosophy, as defined in MIL-PRF-28800, shall be as specified in the contract or order.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commericial market. The government reserves the right to require proof of such conformance.

5.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

5.3 The contracting officer has the option of accepting or rejecting the product.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Military Specifications, Standards, and Handbooks referenced herein may be obtained from the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

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7.1.2 The Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

MILITARY INTEREST

Custodian:
Air Force – 99

Preparing activity:
Air Force - 84

Agent:
Air Force - 99

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